

Full Technical Whitepaper Package

Tier C — Deep Due Diligence · NDA Gated

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What Is In This Package

This Tier C package contains the complete technical record of the ETFT (AutoScientist) project as of April 2026. It is intended for technical due diligence by engineering partners, deep-tech VCs, and research organisations.

Document 1 — Core Whitepaper

File: ETFT.pdf

Evolutionary Trajectory Fine-Tuning: Reverse-Engineering Algorithmic Progression

Towards an Open-Loop Autonomous R&D Ecosystem and General Innovation Accelerator

Lukas Benda, April 2026

Covers:

- Section 1: Introduction — LLM limitations in architectural innovation; OpenAI Parameter Golf challenge
 - Section 2: Theoretical Framework — The Discrete Gradient; fitness function $F(a)$; evolutionary trajectory $T = (a_0 \dots a_n)$
 - Section 3: Methodology — The Core ETFT Ecosystem
 - AI-Assisted Top-Down Regression Pipeline (Figure 1)
 - Generative Trajectory Dataset (D_{Gen})
 - Structural-Rationale Meta-Dataset ($D_{\text{Rationale}}$)
 - Probabilistic Performance Estimator (D_{Perf}) and CI/CD Validation Pipeline
 - Section 4: Addressing the Parametric Blindspot — The Open-Loop Architecture
 - Retrospective Pareto (80/20) Delta Analysis
 - The Agentic Research Loop (ARL): Phase 1 Literature Synthesis (Agentic RAG) + Phase 2 Empirical Micro-Experiments
 - In-Context Reinforcement Learning from negative feedback
 - Figure 2: Open-Loop Empirical ETFT Ecosystem
 - Section 5: The Unified Synthesis — Generating SOTA+1
 - Section 6: Conclusion — The General Innovation Accelerator
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Document 2 — Proof of Concept

File: PoC-ETFT-GPU-Poor.pdf

The "GPU-Poor" ETFT Proof of Concept: Autonomous SOTA+1 Discovery in TinyML Time-Series Classification

Lukas Benda, April 2026

Covers:

- Domain selection rationale (TinyML for fast iteration cycles)
 - Triad architecture: Cognitive Layer (LLM API) + Agentic RAG Layer + Physical Evaluation Layer
 - In-Context Meta-Dataset: evolutionary trajectory $a_0 \rightarrow a_3$ for 1D time-series classification
 - Full Python orchestration code (Listing 1 — Autonomous ETFT Evaluation Loop)
 - Cognitive prompt architecture (Pareto analysis + literature retrieval directives)
 - Empirical results: Accuracy 94.2%, Peak RAM 11 KB — surpasses human SOTA (93%, 18 KB)
 - Reproducibility: runs on consumer RTX 4090 or free Google Colab
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Document 3 — Peer Review Report

File: Concept_LLM.pdf

Official Peer Review Report — ETFT Whitepaper

Track Category: Position Paper / Architectural Proposal · Verdict: Strong Accept (Oral Presentation)

Review highlights:

- Endorsement of AI-Assisted Regression Pipeline and D_Gen / D_Rationale / D_Perf tripartite dataset architecture
 - Assessment of Retrospective Pareto (80/20) Delta Analysis as "the strongest conceptual maneuver in the paper"
 - Validation of Agentic Research Loop as a structurally sound solution to the parametric blindspot
 - Recognition of organic negative data recycling as a native data-asymmetry solution
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Document 4 — Long-Form Technical Paper

File: Evolutionary_Trajectory_Fine-Tuning.pdf

Extended technical treatment of the ETFT framework covering mathematical formalisation, full methodology detail, and extended experimental results.

Document 5 — Visionary Capstone

File: `The Meta-Evolutionary Epoch.pdf`

The Meta-Evolutionary Epoch: Elevating the Universal Innovation System from End-Products to Epochal Paradigms

Lukas Benda, April 2026

Covers:

- Second-Order Innovation: ETFT applied to foundational scientific papers (e.g., predicting post-Transformer architectures)
- Organisational Macro-Algorithms: ETFT applied to human coordination systems with human wellbeing as the loss function
- Meta-Evolution: ETFT recursively applied to improve its own architecture (self-referential improvement loop)
- The Universal Innovation System as a societal-scale acceleration engine

Media Asset

File: `ETFT__AI_Research_Scientist.mp4`

Live demonstration of the ETFT autonomous agent executing a discovery cycle.

Architecture Deep-Dives Available on Request

The following internal technical artefacts are available to qualified technical partners under active NDA and are **not** included in this package:

Artefact	Description
CI/CD Validation Pipeline spec	Full implementation detail of the de-optimisation validation harness
D_Perf training schema	Feature engineering and negative sampling strategy for the Probabilistic Heuristic Filter
ARL orchestration architecture	Agent communication protocol, retry logic, RL reward shaping
Benchmark datasets	Proprietary evolutionary trajectory datasets for DL and TinyML domains

IP Protection Statement

This document package constitutes confidential trade secret material belonging to Lukas Benda / AutoScientist (ETFT) project. The existence and timestamp of this package have been registered with **WIPO PROOF** prior to distribution, establishing a verifiable digital fingerprint and priority timestamp admissible as evidence of prior art.

Unauthorised use of the concepts, architectures, or datasets described herein — including Pareto Delta Analysis, the Probabilistic Heuristic Filter training methodology, and the Organic Negative Data flywheel — may constitute misappropriation of trade secrets under applicable law (including the EU Trade Secrets Directive 2016/943 and US DTSA).

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